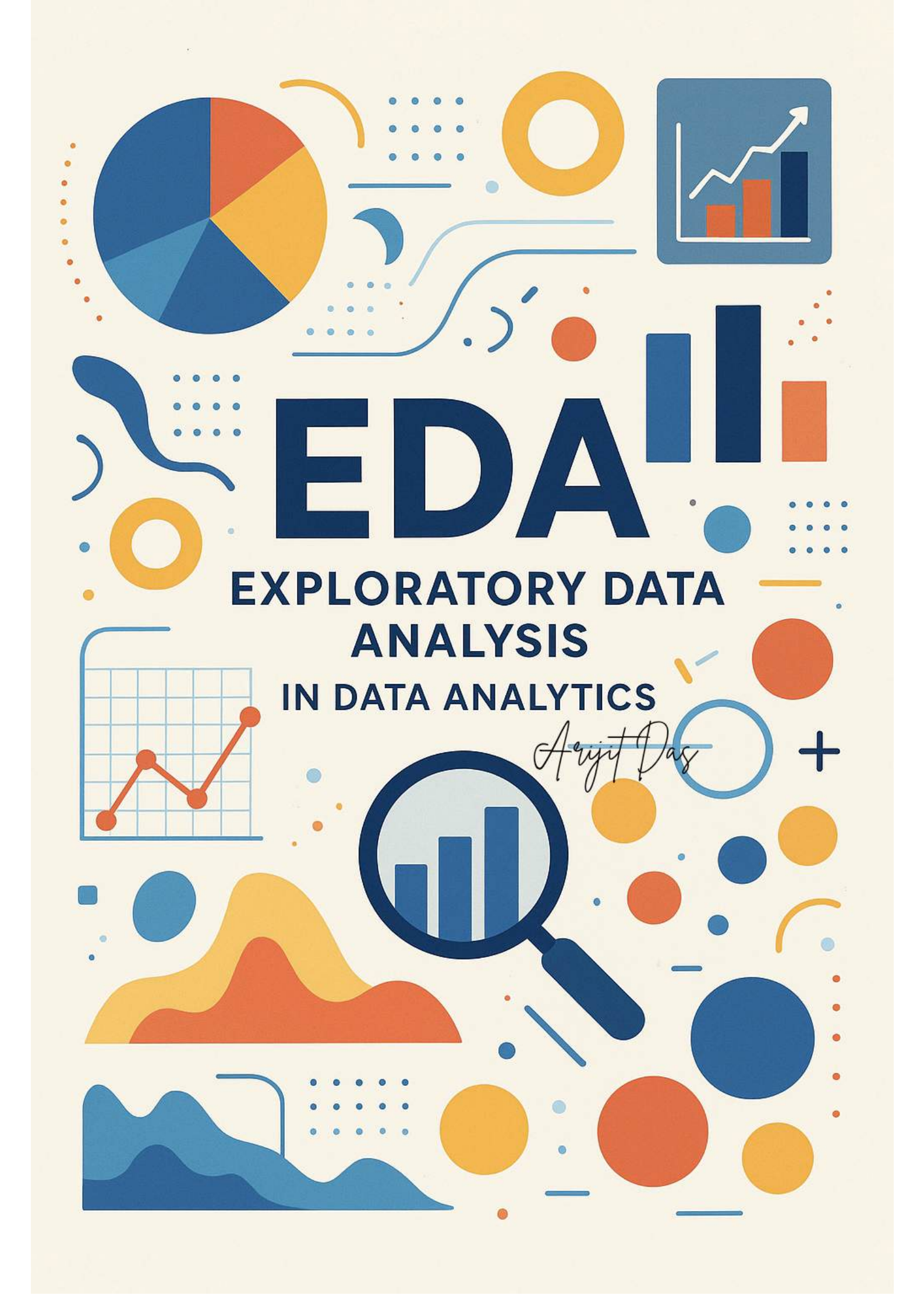




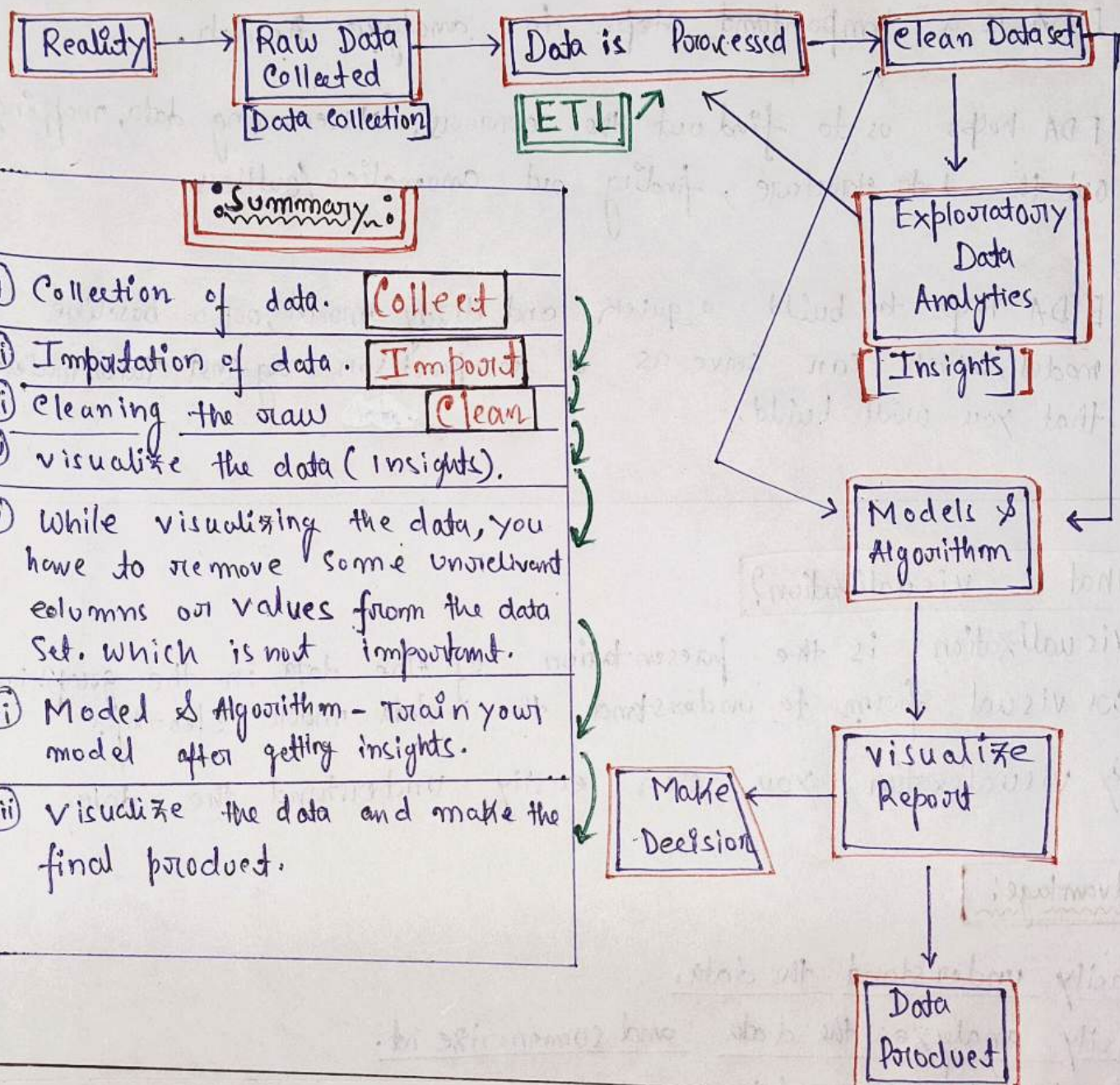
EDA

EXPLORATORY DATA ANALYSIS

IN DATA ANALYTICS

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Data Analytics / Science Process :



1. What is EDA?

⇒ EDA is a method of analyzing the data. Basically it is a set of methods or steps to summarize the characteristics of the dataset.

Aim:

- The main aim of EDA is to obtain confidence in a data to an extent where we are ready to engage a machine learning model.

- EDA is a important steps to analyze the data.

- EDA helps us to find out the errors, discovering data, mapping out the data structure, finding out anomalies/outliers.

- EDA helps to build a quick and dirty model, or a baseline model, which can serve as a comparison against later models that you will build.

② What is visualization?

⇒ visualization is the presentation of the data in the graphical or visual form to understand the data more clearly.

By visualization you can easily understand the data.

■ Advantage:

i) Easily understand the data.

ii) Easily analyze the data and summarize it.

iii) Helps to get meaningful insights.

iv) Helps to find trend or pattern of the data.

■ Tools:

→ Line chart

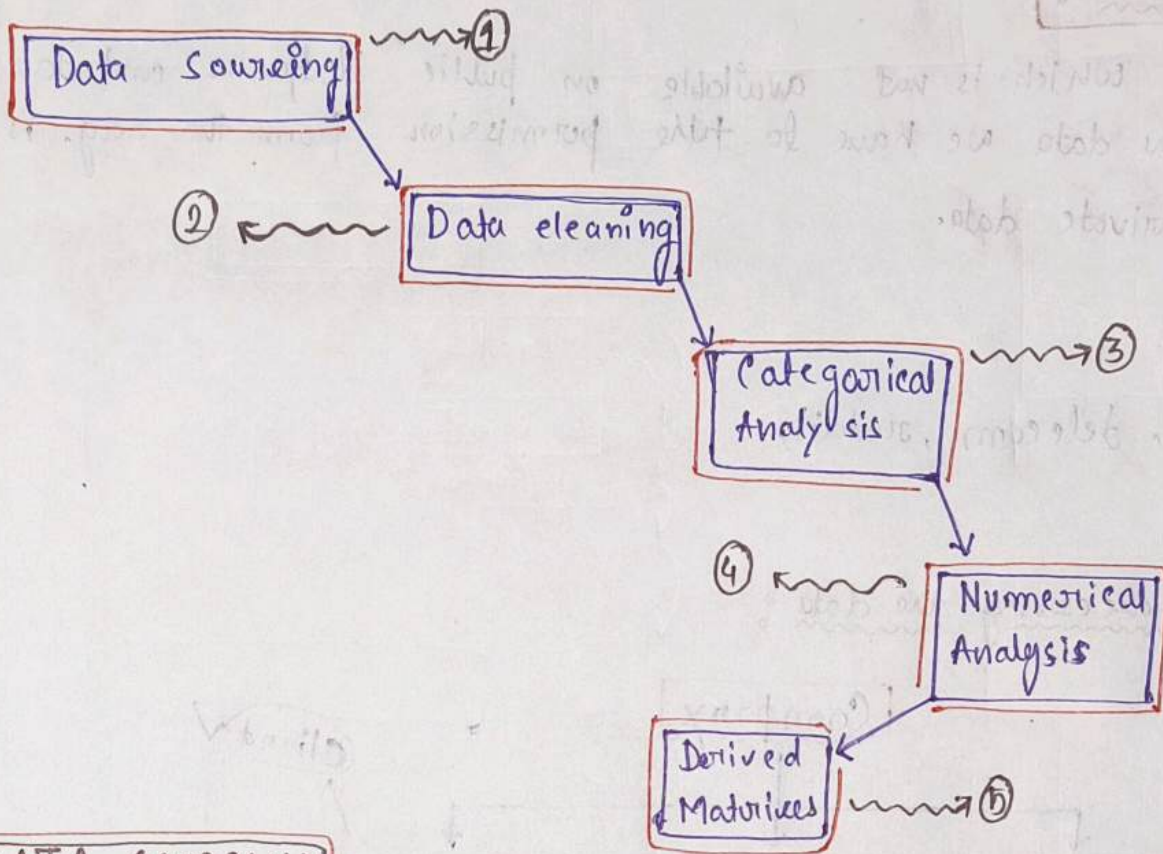
→ Pie plot

→ Bar chart

→ Histogram chart

→ Violin plot

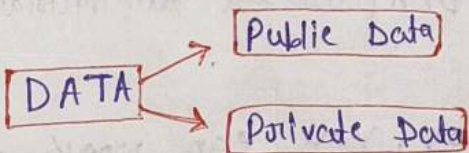
Steps in EDA:



DATA SOURCING

What is data sourcing?

⇒ Data sourcing is a process of gathering the data from many resources as external or internal data collection.



Public data:

The data which is easy to access without taking any permission from the agencies is called Public data.

Example:

Govt. & other Public Sectors.

eg: Kaggle

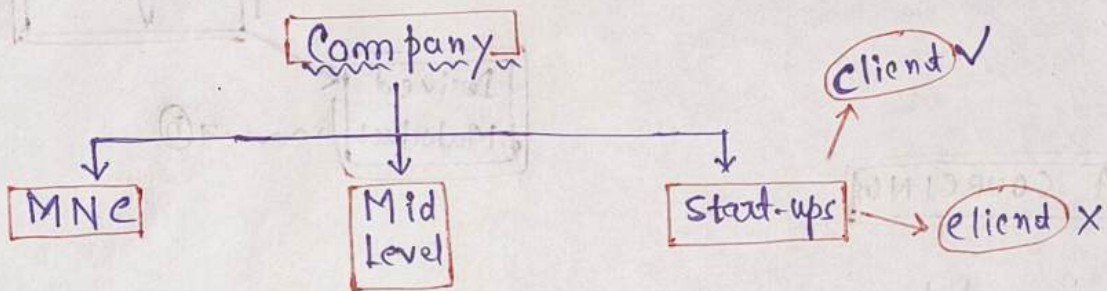
Private Data :

The data which is not available on public platform and to access the data we have to take permission from the org. is called private data.

Example:

Banking, telecom, retail,

Steps of accessing the data :



- *** If you are a data analyst / data scientist in TOP MNEs there are huge amount of databases. To access the databases, you have to grant permission from the DBA (Database Administrators).

→ After granting permission, you can work on your FDA part.

- *** If you are in any mid level company. There is the same process as previous.

- *** If you are working in start-ups.

*** → If the start-ups has database, you have to grant permission from the DBA.

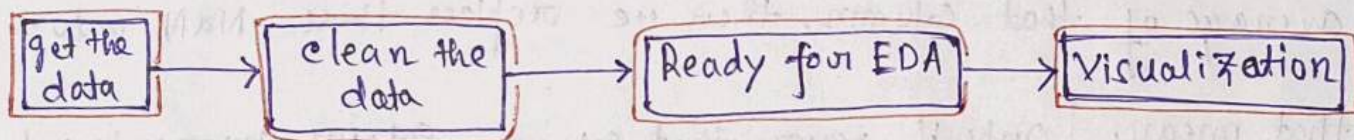
*** → If there is no database, you have to get the data from the govt. or public sectors.

DATA CLEANING:

What is Data cleaning?

After collecting the data, you have to clean the data. By cleaning the data you can easily get rid of those values or information that are not important. ❌❌❌

Data cleaning is a process of clean the data for further better data analysis and to engage with machine learning model.



The benefit of cleaning the data is you can get a fresh and clean dataset, so that you can easily go for EDA or train the ml model.

② Measure steps of data cleaning:

↳ Handle missing values. (NaN) [Not a Number]

↳ Outlier Treatment. [Too maximum or too minimum]

↳ Handle invalid values.

↳ Drop/delete those redundant values {Duplicate}

↳ Standardization of the data.

Handling missing value

① Delete Row / Columns:

If a column contains 75% of missing value, we remove that column entirely.

② Replacing with mean / median / mode:

If you see any few number of rows which contains missing value, we replace it with mean. we first get the average of that column, then we replace those NaN values with that mean. [Only if your that column contains numeric values].

③ Algorithm is Computation:

Some machine learning model supports to handle the missing values. Like KNN, Naïve Bayes, Random forest.

Outlier Treatment :

Outliers are the extreme values which are present in a dataset. Outliers do not fit into the normal behaviour of the data. So you can remove those outliers.

Method to detect outlier:

→ Box plot [IQR $\rightarrow Q_1 \rightarrow Q_3$]

→ Histogram

→ Line chart

→ Scatter plot

→ Z-Score

Some exceptions :

* Some times you don't have to remove outliers, sometimes it is very important information.

* In Day by Day weather data analysis, you have to find out the maximum temperature of a week/month/year.

Example :

Suppose you have a column in your dataset, Salary.

Salary
20000
25000
28000
30000
100000

Now from this,

First step: (Find the outlier)

$$Q_1 = df['Salary'].quantile(0.25)$$

$$Q_2 = df['Salary'].median()$$

$$Q_3 = df['Salary'].quantile(0.75)$$

$$IQR = Q_3 - Q_1$$

2nd step: (Find the range of values you want to see) (except outliers)

$$lower = Q_1 - 1.5 \times IQR$$

$$upper = Q_3 + 1.5 \times IQR$$

3rd step: (Now find out the values)

$$filter_rows = df[(df['Salary'] >= lower) \&$$

$$(df['Salary'] <= upper)]$$

If you take Sample out of many rows:

① Step-1:

$df1 = df['Salary'].sample(10)$

$df1$

② Step-2:

$df1.sort_values(inplace=True)$

$df1$

③ Step-3:

$Q1 = np.percentile(df1, 25)$

$Q2 = np.median(df1)$

$Q3 = np.percentile(df1, 75)$

$IQR = Q3 - Q1$

④ Step-4:

$upper = Q3 + 1.5 * IQR$

$lower = Q1 - 1.5 * IQR$

⑤ Step-5:

$filter_outlier = df1[(df1 >= lower) \& (df1 <= upper)]$

$filter_outlier$

Important Note:

- When we calculate that upper & lower bound in step-4, we try to find out the range of the dataset.
- By doing this, if you see your all data (values) lies between that lower & upper bound, you are safe, you have 0 outliers in your dataset.
- Or if you see, your ~~some~~ few data from lower to upper are out of that lower & upper bound, you can easily understand your dataset has outliers.

Feature Scaling: For predictive analysis:

- Feature scaling is a method to transform the all the data or different range of data into a single scale. It standardize the whole dataset. It is very important for predictive analysis.
- When we are dealing with independent variable where all the data are in different range of values or units or features, then we have to standardize or normalize the variable.

Practical:

Standardization:

$$X' = \frac{X - \mu}{\sigma}$$

- $X \rightarrow$ element value
- $\mu \rightarrow$ Average or mean
- $\sigma \rightarrow$ Standard deviation

Suppose we have a dataset of height or salary.

salary = [10000, 25000, 30000, 60000]

height = [160, 180, 200, 150]

for, predictive analysis the ML model compares one data with another. In that case our model will prioritize the salary first, not the height because the amount of salary or values in that column is very large in comparison to height. So we have to standardize and normalize that two column.

normalize :

$$X' = \frac{X - \min(X)}{\max - \min}$$

$X \rightarrow$ current value

$\min \rightarrow$ minimum value

$\max \rightarrow$ maximum value

Practical

import pandas as pd

import numpy as np

from sklearn.preprocessing import MinMaxScaler(), StandardScaler()

df = pd.read_excel("file.xlsx")

columns = df[['Salary', 'height']]

normalization, standardization.

norm = MinMaxScaler()

std = StandardScaler()

df[['Salary', 'height']] = norm.fit_transform(df[['Salary', 'height']])

df[['Salary', 'height']] = std.fit_transform(df[['Salary', 'height']])

Handle Invalid Value

→ Encode Unicode properly.

→ Convert incorrect types of data.

→ Correct values that goes beyond range.

→ Correct wrong structure

A) Encode Unicode properly:

Encoding in data means the way characters are stored as bytes in a file.

* If pandas can not decode the file with the given encoding, you'll get an error like:

Unicode Decode Error: 'Utf-8' codec can not decode byte 0xe9 in position 10: invalid continuous byte

* How to overcome that thing?

→ Try to read with utf-8 (default):

```
df = pd.read_csv("data.csv", encoding='utf-8')
```

→ If text looks broken, try latin 1 (works most of the time):

```
df = pd.read_csv("data.csv", encoding="latin1")
```

→ still wrong? try this

```
df = pd.read_csv("data.csv", encoding="cp1252")
```

Convert inconsistent data types:

convert the inconsistent data types to the consistent data type.

⊗ Suppose you have a column that seems like it is numeric but most of the time it will present as string or object type.

⊗ Then you have to convert it into numeric or integer.

```
df["col_name"] = df["col_name"].astype(int)
```

⊗ or Sometimes your data can be present as float in object type, then you can get it as integer by

```
df['col_name'] = df['col_name'].astype(float).astype(int)
```

② Correct wrong structure:

Values that don't follow a defined structure can be removed.

* Suppose, you have a column in your dataset, and you notice that column contains ph no. of every user but it is in wrong format. Means a phone num. starting with (+91*****).

You can remove that by,

```
df['ph-num'] = df['ph-num'].astype(str).str[-10:]
```

* Suppose, you have pincode that also contains 12 digits but it have to be in 6 digit.

1st step: [check if it is 6 digit or not]

```
df.loc[df['Pin'].astype(str).str.len() != 6]
```

2nd step: [improve it]

```
df['Pin'] = df['Pin'].astype(str).str[-6:]
```

```
df['Pin'] = df['Pin'].astype(str).str[:6]
```

① Correct values that goes beyond range:

* 1st step:

`df.loc[df['col'] < 0]`

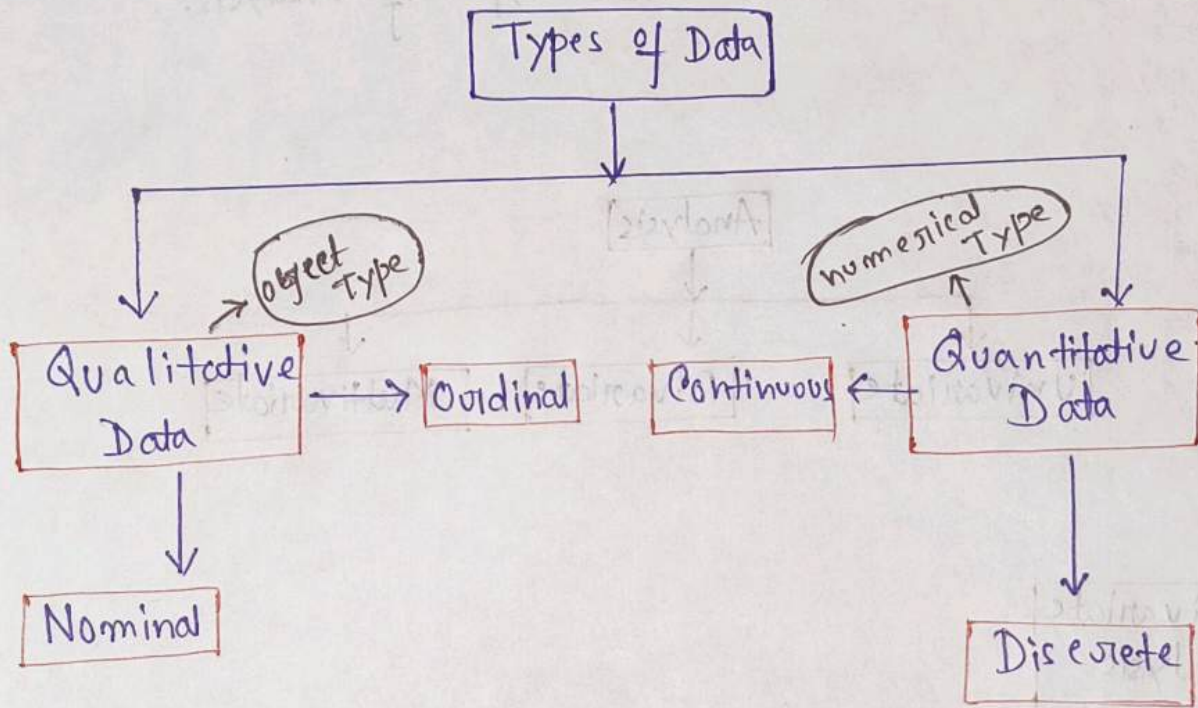
That checks any numeric data are less than 0 or not.

* 2nd step:

`df.loc[df['col'] < 0, 'col'] = np.nan`

After that you can easily fill the nan values with mean or average.

Types of Data:



① Qualitative Data:

The categorical data (gender, product, name) is called Qualitative data.

② Quantitative Data:

The numerical data is called Quantitative data.

① Nominal data: Which data has no order.

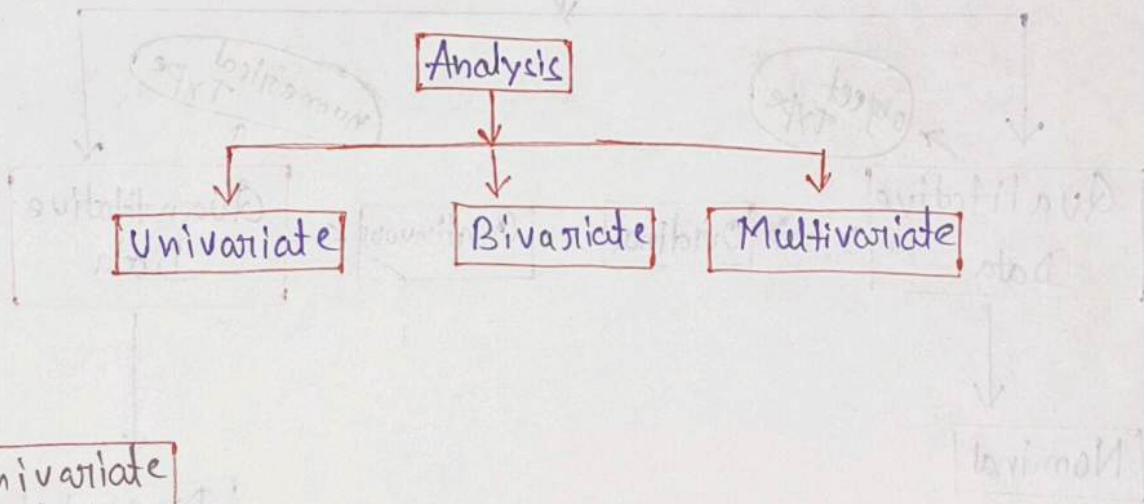
② Ordinal data: Which data has an order

③ Discrete: The numerical data without any decimal part.
[42/50/43]

④ Continuous: The numerical data with decimal part.
[36.5, 40.0]

Types of Analysis

Defn: There is total three types of Analysis.



① Univariate Analysis

* Uni + variate (1) (variable) → working with 1 variable/column in a table.

* works on both type of data.

- ↳ Qualitative
- ↳ Quantitative

A Qualitative: Categorical data.

B Quantitative: Numerical data.

③ Multivariable Analysis:

- ⊗ Data which has more than two variables.
- ⊗ Beyond univariate & Bivariate.

Exception:

- ⊗ If you have enough time, then only you can go for multivariate analysis.
- ⊗ Otherwise, you can only work on uni or bivariate analysis.

② Bivariate Analysis

Bi + variate $\rightarrow$ Working with two variables (columns).
 $\downarrow$ $\downarrow$
2 variable

* works on both type of data.

$\rightarrow$ Qualitative
 $\rightarrow$ Quantitative

① Qualitative: Means the categorical data.

② Quantitative: Means the numerical data.

* CASES

① Categorical $\leftrightarrow$ Numerical

Example: Gender vs salary (Suppose you are calculating each gender's salary. [Male, female].

Tools: Box-plot, bar plot;
(visualization)

② Numerical $\leftrightarrow$ Numerical

Example: Age vs salary (suppose you are calculating each age salary).

Tools: line plot, boxplot, scatterplot;

Numerical Analysis

We perform various analysis over Numerical data.

- ⊛ If you are working with one or two or more variables, you have to calculate the mean, median, mode, Standard Deviation, Variance, min, max, etc.

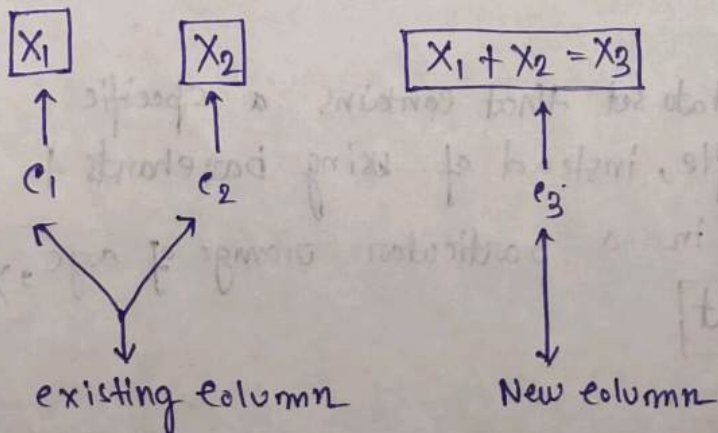
Derived Metrics

Defn:

Derived metrics create a new variable from the existing variable to get a insightful information.

Example:

- Suppose you have a dataset having multiple columns, and you want to add new column then you manually add a new column based on the existing column.
- Like, you have a dataset of sales. You want to add a new column which will contain the average of all sales.
- Means you derived a new column from a existing column



Derived Metrics

Feature Binning

①

Feature Encoding

②

Form-Domain Knowledge

③

calculated from data

④

④ Feature Binning:

It transform continuous or numerical variable into categorical value.

→ Equal width:

Equal width separate the continuous variable into a single range.

→ Equal frequency:

It separates the continuous variable into several categories having approximately same number of values.

④ Example:

Suppose you have a dataset that contains a specific continuous age group of people, instead of using bar charts to show the number of people in a particular range of age, you have to use histogram plot.

③ Calculated from data:

Suppose you want to add a new column that will contain a numerical data. And that data is derived from another column.

* Example:

- Suppose you have a column that contains salary of employees and you want add new column that will contain some bonus salary.
- You have to create new column and then increase the salary bonus for each employees.

Steps to calculate Feature Binning:

1) Equal width:

* Step-1:

Choose a column first.

`df['age']`

* Step-2

`df['age'].max()` # in a variable `max`

`df['age'].min()` # in a variable `min`

* Step-3

Assume a `bin = 4 (K)` `K=4`

$\text{width} = (\text{max} - \text{min}) / K$

* This is the width between all ages range.

`df['equal-width'] = pd.cut(df['Age'], bins=4)`

* Step-4

plot the graph by histogram.

```
sns.histplot(x='age', bins=4, kde=30)  
plt.show()
```

* Step-5

Also you can get to know the total ages that falls into each bins.

```
df['equal-width'].value_counts()
```

* NOTE

→ How you can get to know how many bins you have to set for your dataset?

* It depends on number of rows present in the dataset.

if,

rows = 500 ↓ (lower than 500), bins = 3-5

rows = (500-1000), bins = 10-20

rows = 5000 ↑ (higher than 5000), bins = 10-20

Feature Encoding

Defn:

Feature Encoding means, when we assign some numerical value to a categorical value which are present in a particular column.

Types:

Label encoding → ①

↓
One-hot encoding → ②

① Label- Encoding:

• Suppose, There is a column called 'Gender'. Which contains Female & Male value only.

• And we are assigning 1 for the Male category and 0 for the female category.

	<u>Gender</u>	<u>New-col</u>
0	Male	1
1	Male	1
2	Female	0
3	Female	0
4	Female	0
5	Male	1
6	Female	0
7	Female	0

② One-hot encoding:

- In this encoding technique, it only works on nominal or non-order type values (categorical - x-axis).
- It creates K -different columns for each categorical value.
- For each category it assigns 1. and rest of the columns assigns with 0.

<u>Location</u>	<u>loc-India</u>	<u>loc-Japan</u>	<u>loc-Nepal</u>
India	1	0	0
Japan	0	1	0
Nepal	0	0	1
India	1	0	0
Nepal	0	0	1
Japan	0	1	0
Japan	0	1	0
India	1	0	0
Nepal	0	0	1
Japan	0	1	0
India	1	0	0